T3 TASK SUBMISSION

HADOOP TRAINING

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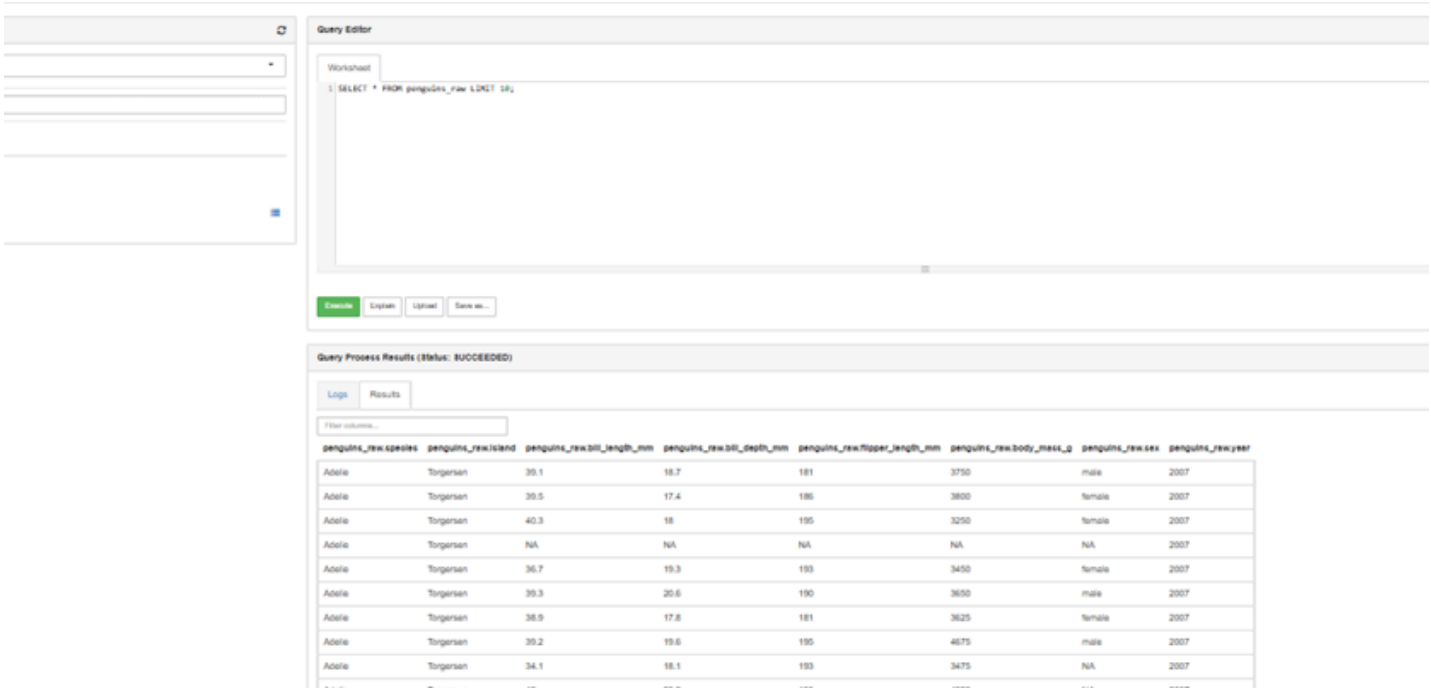
Section 1. HDFS Using the Command Line

- wget Hadoop Training.
- ls
- hdfs dfs -ls
- hdfs dfs -mkdir T3Task
- hdfs dfs -copyFromLocal penguins.csv T3Task/penguins.csv
- hdfs dfs -ls T3Task

```
maria_dev@sandbox-hdp ~]$ hdfs dfs -ls T3Task
ound 1 items
rw-r--r--   1 maria_dev hdfs      15241 2026-10-17 11:51 T3Task/penguins.csv
```

Section 2. Hive Table and HiveQL

- Run SELECT * FROM penguins_raw LIMIT 10;



- **Count records for each penguin species and island.**

```
SELECT
    penguins_raw.species,
    penguins_raw.island,
    COUNT(*) as Total
FROM penguins_raw
GROUP BY penguins_raw.species, penguins_raw.island;
```

Query Process Results (Status: SUCCEEDED)

Logs | Results

Filter columns...

penguins_raw.species	penguins_raw.island	total
Adelie	Biscoe	44
Adelie	Dream	56
Adelie	Torgersen	52
Chinstrap	Dream	68
Gentoo	Biscoe	124

- **Find average body mass and average flipper length for each species.**

```
SELECT
   penguins_raw.species,
   AVG(penguins_raw.body_mass_g) as body_mass,
   AVG(penguins_raw.flipper_length_mm) as
flipper_length
FROM penguins_raw
GROUP BY penguins_raw.species;
```

Query Process Results (Status: SUCCEEDED)

Logs | Results

Filter columns...

penguins_raw.species	body_mass	flipper_length
Adelie	3700.662251655629	189.95364238410596
Chinstrap	3733.0882352941176	195.8235294117647
Gentoo	5076.016260162602	217.1869918699187

- **Count missing values in bill_length_mm, flipper_length_mm, body_mass_g, and sex.**

```
SELECT
   SUM(CASE WHEN penguins_raw.bill_length_mm IS NULL THEN 1 ELSE 0 END) AS missing_bill,

   SUM(CASE WHEN penguins_raw.flipper_length_mm IS NULL THEN 1 ELSE 0 END) AS missing_flipperlength,

   SUM(CASE WHEN penguins_raw.body_mass_g IS NULL THEN 1 ELSE 0 END) AS missing_bodymass,

   SUM(CASE WHEN penguins_raw.sex IS NULL THEN 1 ELSE 0 END) AS missing_gender
FROM penguins_raw;
```

Query Process Results (Status: SUCCEEDED)

Logs | Results

Filter columns...

missing_bill	missing_flipperlength	missing_bodymass	missing_gender
0	0	0	0



Section 3. ELT with WebHDFS InsecureClient

- Using Jupyter Notebook on Windows, connect to the Hadoop sandbox through WebHDFS using Python InsecureClient and load penguins.csv from HDFS into a Pandas DataFrame. Analyze its structure and missing values.

Structure

```
Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    str
 1   island             344 non-null    str
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    str
 7   year               344 non-null    int64
dtypes: float64(4), int64(1), str(3)
memory usage: 27.6 KB

Dataset shape: (344, 8)
```



Missing Values

```
Missing values:
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64

Missing values in selected columns:
bill_length_mm        2
flipper_length_mm     2
body_mass_g           2
sex                  11
dtype: int64
```

Section 3. ELT with WebHDFS InsecureClient

- Clean and preprocess the data by handling missing values, converting body mass to kilograms

```
Missing values before cleaning:
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64

Missing values after cleaning:
species              0
island               0
bill_length_mm       0
bill_depth_mm        0
flipper_length_mm    0
body_mass_g          0
sex                  0
year                 0
body_mass_kg         0
dtype: int64
```

- Body Mass

Body mass in kilograms:

	species	body_mass_g	body_mass_kg
0	Adelie	3750.0	3.75
1	Adelie	3800.0	3.80
2	Adelie	3250.0	3.25
4	Adelie	3450.0	3.45
5	Adelie	3650.0	3.65

Section 4. PySpark Application on Hadoop

- Develop and run a PySpark application (spark/penguins_spark.py) using spark-submit to read penguins_clean.csv from HDFS and display its schema.

```
Penguins Dataset Schema:
root
 |-- species: string (nullable = true)
 |-- island: string (nullable = true)
 |-- bill_length_mm: double (nullable = true)
 |-- bill_depth_mm: double (nullable = true)
 |-- flipper_length_mm: double (nullable = true)
 |-- body_mass_g: double (nullable = true)
 |-- sex: string (nullable = true)
 |-- year: integer (nullable = true)
 |-- body_mass_kg: double (nullable = true)
```

- Select relevant columns, filter records with flipper length below or equal to zero, apply a simple filter or sorting operation, and cache the results.

```
+-------+---------+--------------+-------------+-----------------+-----------+------+----+------------+
|species|   island|bill_length_mm|bill_depth_mm|flipper_length_mm|body_mass_g|   sex|year|body_mass_kg|
+-------+---------+--------------+-------------+-----------------+-----------+------+----+------------+
| Adelie|Torgersen|          39.1|         18.7|            181.0|     3750.0|  male|2007|        3.75|
| Adelie|Torgersen|          39.5|         17.4|            186.0|     3800.0|female|2007|         3.8|
| Adelie|Torgersen|          40.3|         18.0|            195.0|     3250.0|female|2007|        3.25|
| Adelie|Torgersen|          36.7|         19.3|            193.0|     3450.0|female|2007|        3.45|
| Adelie|Torgersen|          39.3|         20.6|            190.0|     3650.0|  male|2007|        3.65|
+-------+---------+--------------+-------------+-----------------+-----------+------+----+------------+
only showing top 5 rows
```



Section 4. PySpark Application on Hadoop

- 1.Save the output as JSON to /user/maria_dev/spark/output/penguins in HDFS, partitioned by a suitable column. Stop the Spark session properly.